

Mehdi Molaei

Senior Staff Engineer, Computational Imaging

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Education

Johns Hopkins University, Baltimore, MD 2012 – 2015

Ph.D., Mechanical Engineering

Advisor: Prof. Jian Sheng, Prof Joseph Katz

Texas Tech University, Lubbock, TX 2012 – 2015

Ph.D., Mechanical Engineering

Advisor: Prof. Jian Sheng

Dissertation: *Hydrodynamics of bacterial locomotion near interfaces.*

University of Minnesota, Minneapolis, MN 2010 – 2012

M.S., Aerospace Engineering and Mechanics

Advisor: Prof. Jian Sheng

Dissertation: *Hydrodynamics of bacteria and solid-surface interaction using microfluidics and digital holographic microscopy.*

Sharif University of Technology, Tehran, Iran 2007 – 2010

M.S., Mechanical Engineering

Advisors: Prof. Saeed Saeedi and Prof. Bahareh Dehghan Firoozabadi

Dissertation: *One-dimensional modeling of the cardiovascular system.*

Sharif University of Technology, Tehran, Iran 2003 – 2007

B.S., Mechanical Engineering

Research Interests

Spatiotemporal modeling and inference; multimodal signal fusion and representation learning; rare-event and rare-class inference; machine-learning-enabled scientific instrumentation; real-time scientific computing; computational imaging and quantitative microscopy; soft and living matter physics; diffusion models.

Professional Appointments and Research Experience

Senior Staff Engineer, Countable Labs, Palo Alto, CA Dec 2023 – Jan 2026

Team Lead, Countable PCR Analysis Software

Led the architecture and delivery of the production analysis platform converting multichannel light-sheet 3D PCR imaging into calibrated single-molecule counts; recognized with the #1 Analytical Scientist Innovation Award (2025).

- Architected and owned an end-to-end ETL and computational imaging pipeline for DNA molecule counting, including 3D reconstruction, artifact and aberration correction, compartment segmenta-

tion, failed-region detection, and unsupervised compartment-state assignment.

- Designed and built an on-premises gRPC client–server analysis system with parallel compute execution, DAG-based dependency scheduling, and asynchronous status reporting, reducing latency from batch turnaround times exceeding 12 hours to live results.
- Led development of a DNA linkage-detection module combining multichannel alignment, colocalization, regression modeling, and training/validation, achieving 97% accuracy with less than 4% false-positive rate.
- Developed and deployed quality-control models for anomaly and defect detection in real-world PCR runs, achieving 75–95% recall at false-positive rates of 5% or below across multiple failure modes.
- Co-led translational research for the Countable 10 system, expanding multiplexed imaging from 4 to 10 channels through channel-design optimization, spectral modeling, and advanced inference methods.
- Developed a 10-color decoding workflow and a multimodal variational autoencoder for combinatorial DNA labeling, enabling scaling from 10-plex to 24-plex at 98% accuracy.

Staff Engineer, Countable Labs, Palo Alto, CA

Aug 2022 – Dec 2023

Tech Lead, Image and Data Analysis Pipeline

Led development of production analysis methods that transformed PCR workflows from manual thresholding and operator-dependent interpretation into automated, quantitative inference.

- Owned end-to-end release execution, including testing, build and release, verification and validation, and deployment of early-access instrument software to external customers.
- Built and shipped a gradient-boosted decision-tree model for ultra-rare molecular detection, supporting sensitivity down to 0.004% minor allele frequency with in-run calibration for drift and assay variability correction.
- Developed and deployed 3D sample-region detection methods that reduced dead volume to less than 5% and preserved valuable DNA samples.
- Built a physics-informed regression calibration model for molecular density estimation, improving precision by 80% and achieving coefficient of variation below 1%.
- Developed data analysis pipelines for multiplex detection, including data cleanup, augmentation, UMAP embedding, clustering, regression modeling, and visual quality-control dashboards, supporting development of a 22-target comboplex panel.
- Defined and monitored key instrument quality metrics, implemented automated magnification calibration, and devised a PCA-based autofocus solution for instrument commissioning.

Research Scientist, University of Chicago, Chicago, IL

Aug 2020 – Aug 2022

Pritzker School of Molecular Engineering and James Franck Institute

- Developed a domain-general spatiotemporal correlation framework for tensor fields in stochastic systems, extending correlation analysis to incorporate internal headings and orientations and enabling robust inference under chaotic dynamics.
- Recast correlation analysis as a data-driven, ensemble-averaged material response function and applied it to actomyosin active systems ranging from model systems to living cells.
- Mentored doctoral students in quantitative imaging and data modeling, coauthored peer-reviewed publications, and presented research outcomes at conferences.

Postdoctoral Researcher, University of Pennsylvania, Philadelphia, PA

Sep 2015 – Aug

2020

Physical Sciences Oncology Center; Crocker Lab; Stebe Lab

- Developed correlated displacement velocimetry, a correlation-based, ensemble-averaged flow estimator built on particle tracking, enabling reconstruction of nanoscale interfacial velocity fields.
- Built a polarization-resolved dark-field optical metrology system for sub-degree gold nanorod orientation tracking and developed physics-informed forward models to infer membrane mechanical parameters in model and cellular systems.
- Designed and built a miniature 3D-printed Langmuir trough enabling simultaneous imaging tensiometry and interfacial microrheology across viscosities spanning approximately 10^{-9} to 10^{-1} N s m $^{-1}$.
- Led studies of bacteria at fluid interfaces, combining physical and biological experiments to quantify interfacial film formation and bacterial motility; mentored doctoral students and contributed invited talks and publications.

Research Assistant, Texas Tech University, Lubbock, TX

Jul 2012 – Aug 2015

Laboratory for Advanced Energy, Bio-Fluids and Imaging

- Built an inline holographic microscopy platform and denoising and 3D-tracking algorithms for bacteria in dense suspensions.
- Developed theoretical and numerical models for bacterial locomotion near solid and fluid interfaces.

Research Assistant, University of Minnesota, Minneapolis, MN

2010 – 2012

Jian Sheng Lab

Conducted experimental and computational studies of bacterial hydrodynamics and locomotion near interfaces, including digital holographic microscopy and microfluidic systems.

Lab Manager, Sharif University of Technology, Tehran, Iran

2009 – 2010

Laboratory of Computational Biofluid Mechanics

Worked on mathematical modeling of blood flow in the cardiovascular system and supported laboratory research activities in computational biofluid mechanics.

Research Engineer, SITEC Co., Tehran, Iran

2007 – 2010

Engineering role in applied mechanical systems and analysis.

Publications

Patent

1. **Molaei, M.** et al. *High dynamic range digital quantitation of targets*. U.S. Patent Application US20250264395A1, Countable Labs.

Peer-Reviewed Research Articles

1. Chauhan, R., Usman, H., Minocha, N., **Molaei, M.**, et al. "Predictive modeling and experimental validation of magnetophoretic delivery of magnetic nanocultures." *ACS Materials Letters*, 2025.
2. Usman, H., **Molaei, M.**, House, S. D., et al. "Magnetically responsive nanocultures for direct microbial assessment in soil environments." *Science Advances*, 2025.
3. Redford, S. A., Colen, J., Zemsky, S., **Molaei, M.**, et al. "Motor crosslinking augments elasticity in active nematics." *Soft Matter*, 2024.
4. Chou, W. H., **Molaei, M.**, et al. "Limiting pool and actin architecture controls myosin cluster

- sizes in adherent cells.” *Biophysical Journal*, 2024.
5. Crane, B. L., Jimenez-Angeles, D., Wang, J., **Molaei, M.**, et al. “Interfacial rheology of lanthanide binding peptide surfactants at the air–water interface.” *Soft Matter*, 2024.
 6. Deng, J., **Molaei, M.**, et al. “Swimmers at interfaces enhance interfacial transport.” *Soft Matter*, 2024.
 7. **Molaei, M.** et al. “Measuring response functions of active materials from data.” *Proceedings of the National Academy of Sciences*, 2023.
 8. Rodriguez-Cruz, P., **Molaei, M.**, Crocker, J. C., et al. “Experimental observations of fractal landscape dynamics in a dense emulsion.” *Soft Matter*, 2023.
 9. **Molaei, M.**, Sreeja, K. K., Graber, Z. T., et al. “Probing lipid membrane bending mechanics using gold nanorod tracking.” *Physical Review Research*, 2022.
 10. **Molaei, M.**, Chisholm, N. G., et al. “Interfacial flow around Brownian colloids.” *Physical Review Letters*, 2021.
 11. **Molaei, M.** and Crocker, J. C. “Interfacial microrheology and tensiometry in a miniature, 3-D printed Langmuir trough.” *Journal of Colloid and Interface Science*, 2020.
 12. **Molaei, M.**, Deng, J., Chisholm, N. G., et al. “Motile bacteria at oil–water interfaces: *Pseudomonas aeruginosa*.” *Langmuir*, 2020.
 13. **Molaei, M.**, Vaccari, L., Leheny, R. L., and Stebe, K. J. “Cargo carrying bacteria at interfaces.” *Soft Matter*, 2018.
 14. **Molaei, M.**, Atefi, E., and Crocker, J. C. “Nanoscale rheology and anisotropic diffusion using single gold nanorod probes.” *Physical Review Letters*, 2018.
 15. **Molaei, M.** and Sheng, J. “Succeed escape: Flow shear promotes tumbling of *Escherichia coli* near a solid surface.” *Scientific Reports*, 2016.
 16. **Molaei, M.**, Barry, M., Stocker, R., et al. “Failed Escape: Solid Surfaces Prevent Tumbling of *Escherichia coli*.” *Physical Review Letters*, 2014.
 17. **Molaei, M.** and Sheng, J. “Imaging bacterial 3D motion using digital in-line holographic microscopy and correlation-based denoising algorithm.” *Optics Express*, 2014.

Review Articles

1. Deng, J., **Molaei, M.**, et al. “Active colloids on fluid interfaces.” *Current Opinion in Colloid and Interface Science*, 2017.
2. Rose, E., **Molaei, M.**, Boyle, J., et al. “Particle tracking of nanoparticles in soft matter.” *Journal of Applied Physics*, 2020.
3. Niepa, T. H. R., Vaccari, S., **Molaei, M.**, et al. “Films of bacteria at interfaces.” *Advances in Colloid and Interface Science*, 2017.

Preprint

1. Lai, J. H., Keum, J., Lee, H. G., **Molaei, M.**, et al. “New realm of precision multiplexing enabled by massively-parallel single molecule UltraPCR.” *bioRxiv*, 2023.

Mentoring and Teaching

- Mentored Ph.D. students and undergraduate researchers at the University of Pennsylvania and the University of Chicago on quantitative imaging, soft matter, and data modeling projects.

- Contributed to undergraduate and teacher-focused summer research programs, including REU and RET activities.
- Teaching Assistant, Experimental Fluid Mechanics, Texas Tech University (Spring 2013).
- Teaching Assistant, Biofluid Mechanics, Sharif University of Technology (Spring 2009).

Selected Conference Presentations

- 1.**Molaei, M.** “Surfactants and fundamental flow fields at interfaces.” APS Division of Fluid Dynamics Meeting, Phoenix, AZ, 2021.
- 2.**Molaei, M.**, Chou, W., Redford, S. A., and Gardel, M. “Local intracellular flow fields driven by active stresses of molecular motors.” APS Division of Fluid Dynamics Meeting, Phoenix, AZ, 2021.
- 3.**Molaei, M.**, Deng, J., Chisholm, N. G., Crocker, J. C., and Stebe, K. J. “Displacement fields around thermally forced and active colloids at interfaces.” AIChE Annual Meeting, Orlando, FL, 2019.
- 4.**Molaei, M.** and Crocker, J. C. “Motion of gold nanorods on the fluid lipid bilayer and cell plasma membranes: interfacial nanorheology.” APS Division of Fluid Dynamics Meeting, Atlanta, GA, 2018.
- 5.**Molaei, M.**, Vaccari, L., Leheny, R. L., and Stebe, K. J. “Cargo carrying bacteria at interfaces.” ACS Colloid & Surface Science Symposium, State College, PA, 2018.
- 6.**Molaei, M.** and Sheng, J. “Motion of bacteria near a solid surface under shear flows.” APS Division of Fluid Dynamics Meeting, San Francisco, CA, 2014.

Professional Service and Activities

- Journal reviewer for *Science Advances*, *PNAS*, *Soft Matter*, *Journal of Fluid Mechanics*, *Langmuir*, and *Nano Letters*.
- Member of the American Physical Society and the American Institute of Chemical Engineers.
- Personal project: *Trajkit*, an open-source Python toolkit for analyzing spatiotemporal trajectories and structured time-series data.

Honors and Recognition

- #1 Analytical Scientist Innovation Award, Countable Labs (2025).
- Recognized graduate student as a GoMRI Scholar for oil-spill research (2014).
- Ranked among the top 0.1% of approximately 10,000 participants in the nationwide entrance examination for M.S. study in Iran (2007).
- Ranked among the top 0.1% of approximately 500,000 participants in the nationwide entrance examination for B.S. study in Iran (2003).